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Network System (ESOFT Metro Campus)

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FC6P01ES Project

Interim Report

Home Automatic Lightning System and Automatic Water Control.

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Module: FC6P01 **Deadline: 02/15/2022**

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Abstract

Internet of Thing (IoT) is trending up and Home Automation is becoming many people need nowadays. The automation system is used to reduce the human effort and make work effective and efficient than a manual system. As technology has become the part of the life so that this project was proposed to make Home Automation System through IoT. So, this system will be developed to make reliable home automation system in which people can trust. In this system, the main focus is concentrated among lightening system and water control which will help to save energy and loss of user and unnecessary burden. While carrying out this project developer has considered many methodologies. Regarding the structure of the project, Incremental Model was finalised by the developer. Along with this lots of research on the similar project were carried out. Analysing the similar project and using its feature it had become used to design the system. After the completion of this project, people can run their household activity automatically with the help of internet.

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1. Introduction

The Internet of Thing which is also known as IoT in abbreviation. It is the concept of Machine to Machine interaction by using the internet among distributed ubiquitous devices in a standardized manner. [CITATION IEE15 \l 3081] IoT has become the hot topic nowadays in the field of Information and Technology although it is not a new concept. It was introduced by Kevin Ashton in early 2000's at MIT's AutoID lab. [CITATION Dun15 \l 3081] IoT is used to sense and collect data from the environment using different sensors. Those data are filtered and been used to carry different activity using its information. After that, lots of research are being carried out in the field of IoT. Similarly, using the concept of IoT the developer is building a Home Automation System Using Raspberry Pi in this project.

1.1 Current Scenario

Home Automation users are increasing on a daily basis, whereas there are 3.5 billions of sensors at 2013 [CITATION ICI15 \l 3081] and is expected to increase to trillion within the next decade. Currently, light, door and security are major features carried by others product. As regards the result that Google just aforementioned Nest Learning Thermostat for $3.2 billion [CITATION MAR14 \l 3081] prove that there is great need of Home Automation in the present context of the world.

1.2 Aim and Objective

The goal of this project is to design a home automation system using the principle of IoT and hardware used is Raspberry Pi. There will a web app in this project which can be used to control the different devices connected to network as per the requirement of user but most of the activities are automated. This system can be accessed from any device that can be connected to the internet.

The objectives of this project are classified as:

a. Research and survey will be conducted for the project. The outcome of research and survey will be analysed scientifically which will help to start the project. Then after, research and requirement analysis document will be produced in 5 weeks.

b. The design and development of the system will be carried as per requirement analysis report with an incremental model. The first system will be ready and also will be tested completely which will be ready to use by end users which need 8 weeks to complete.

c. Again, design and development of remaining part of the system will be carried out followed by requirement analysis report. Then another system is also completed. Then this system is also tested and make ready to use by end users which need 8 weeks to complete.
d. Both systems will be merged and final system will be developed than that system will be reviewed by the end users and that feedback will be analysed for further work to make the final product. This task requires 2 weeks to complete.
e. Analysis of feedback will be done and possible work will be carried out according to feedback which will improvement the system than before. It requires 3 weeks to complete.
f. The system is deployed as automated home system. Along with the deployment of the system other work like making user manual, and technical documentation will be completed. It requires 2 weeks to complete all this work.

1.3 Motivation

In the recent research of Icontrol Network, it was found that most of the people wants the smart devices that will automate themselves. They have also conducted a survey in the smart home which has the finding that 50% of the American population will buy at least one connected device by next year. Assuming this we can also say that IoT and device for home automation is going to go on a global market. Along with Icontrol Network, there was another survey piloted by Gartner and they have found that by 2022, a typical home will contain more than [CITATION Ico15 \l 3081]

With the finding of Icontol and Gartner, the developer has come to the conclusion that IoT and home automation system is going to develop to next level within few years which has motivated him to work on this project. With his own survey result which has outcome that people are euthanasia's about home automation system and is ready to use if it is relabel and cost efficiency (Annex A).

1.4 Limitation and Scope

As home automation system can be a big project but due to some limitation and scope developer has narrowed down this project in following points on basis of limitation and scope:

a. Facility will be only in one room at the beginning then further work is also planned to work with a concept of pervasive computing.

b. This project will have Raspberry Pi as a hardware and others sensor for the automation process which includes the motion sensor and photoresistor (Light dependent resistors).

c. In any case, if a device is need to be used manually then there will a web application which will be hosted on Heroku for free in a cloud which will enable developers to build, run, and operate applications easily.

d. As being the project in IoT there will a great role of the internet.

1.5 Structure of Report

1.5.1 Background

This report includes other subtopic like background as chapter 2 where technical and non-technical things of system is defined in brief. It also includes the similar project which is related to do this project.

1.5.2 Development

In chapter 3, there is development where methodology considered is described and chosen methodology phases is defined. The current engagement of project is also defined in this chapter.

1.5.3 Progress Review

In chapter 4, there is progress review and further work for the project is defined along with Gantt chart in this chapter.

1.5.4 Conclusion

In chapter 5, the conclusion of the project is summarised.

2. Background

2.1 Project Overview

An internet based home automation system is used to control the home from remote or within the home. IoT and Home Automation System is trending over the world and there are lots of projects based on these. In the context of Nepal, while doing research and also looking at the real life scenario there is not Home Automation but people are looking for it. After completion of this project a certain level of needs will be fulfilled and it that further enhancement will be done as due to time constraint, all features cannot be covered in this project. [CITATION RAN16 \l 3081]

This system will include Raspberry Pi which will control all the sensors to make Home Automation. As well as there will be a web application that will be used to control manually.

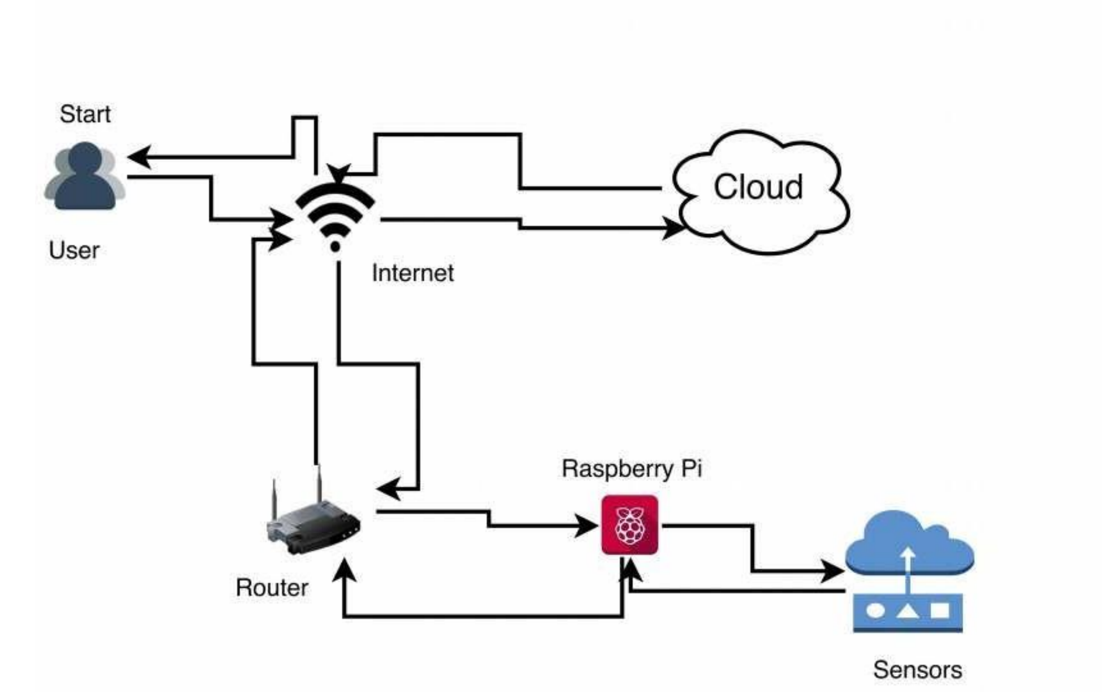

Figure 1: Network Architecture

As network structure in Figure 1 at first user will import their requirement for automation and then system will be independent of user until external interfere is done. In this system, the user can control the system as the data is stored in the cloud as a web application through the internet. Then all the control of sensor and internet is connected through the router. As IoT is a machine to machine interaction all the component in this system will be inter link to each other directly or indirectly.

2.2 Similar Projects

a. leakSMART

leakSMART a trademark of Waxman Consumer Products Group, Inc. leakSMART monitor, detect and control the water leakage. [CITATION lea16 \l 3081]

Figure 2: leakSMART [CITATION lea16 \l 3081]

This system detects water leakage at home and shuts the water supplies within 5 seconds which will prevent the home from getting flooded. This is also an IoT project that can also be controlled remotely. [CITATION lea16 \l 3081]

b. Belkin WeMo

Belkin Wemo is home automation system which is used to monitor and control WeMo-based switches and plug.

Figure 3: Belkin WeMo [CITATION WeM16 \l 3081]

This will control the LED light bulbs and motion sensor from web app and smartphone. This project does not require any hub to control, where everything is managed through Belkin's own free cloud service. Along with this it also supports other small appliances which has low-voltage DC switch and sensor into WeMo application can remotely control those devices. [CITATION Bel16 \l 3081]

c. Connected by TCP

TCP's home lighting automation system controls the lighting system remotely and wirelessly which includes a gateway device that is plugged into the home router for wireless remote control.

Figure 4: Connected by TCP [CITATION Ama16 \l 3081]

It is supported in iOS, Android and has a web application to control lighting system. This project has features like dimming on demand which will allow the user to choose brightness of light. In this project, the bulb offered is A-19 and BR-30 compatible LED bulb that can support up to 250 lamps and control them individually. [CITATION TCP16 \l 3081]

2.3 Comparison Table

Analysing these project, they have some similarity like they all are home automation system and all of them are based on IoT. Although there is more dissimilarity than similarities. Each and every project have their own features and uniqueness that can help the developer to take an idea about the different concept and work out with it.

Features	leakSMART	Belkin WeMo	Connected by TCP	This Project
Control Remotely	Yes	Yes	Yes	Yes
Platform Independent (Web App)	Yes	Yes	Yes	Yes
Android App	Yes	Yes	Yes	No
iOS App	Yes	No	Yes	No
Water Control System	Yes	No	No	Yes

Light Control System	No	Yes	Yes	Yes
Multiple Sensors	No	Yes	No	Yes

Table 1: Comparison Table

The developer has not considered to make iOS and Android app as there will be web application which is platform in dependence. So, currently iOS and Android app does not seem to be required.

2.4 Further Enhancement

Home Automation is a big project and can be extended to any level in the present scenario. All the manual work carried out in our home currently can be atomised to some level. With research and technologies trending up developer has decided to change IoT Home Automation System into Pervasive Computing Model where each and every device of home will have microchip on it and they will interact with each other without human interaction and which will also have self-decision capacity.

3. Development

Software development methodology (SDM) technique used by the developer to develop quality software in systematic order. There are different SDM which is based on the experience of different other developers and the assumption made by them. The different SDM has its own strength and weakness. So, choosing a software engineering methodology is also one of the toughest parts of the development as choosing the wrong methodology may lead to the failure of the project. So, looking at the requirement of the project developer has research on different methodologies and has selected the one best methodology that can be used to complete this project successfully.

3.1 Methodology Selection

3.1.1 Software Prototyping Model

Software prototyping model is the process of developing software by making a different model with limited functionality, but the prototypes will not be exactly same as the original software. While doing this project the developer has considered software prototyping methodology at first.

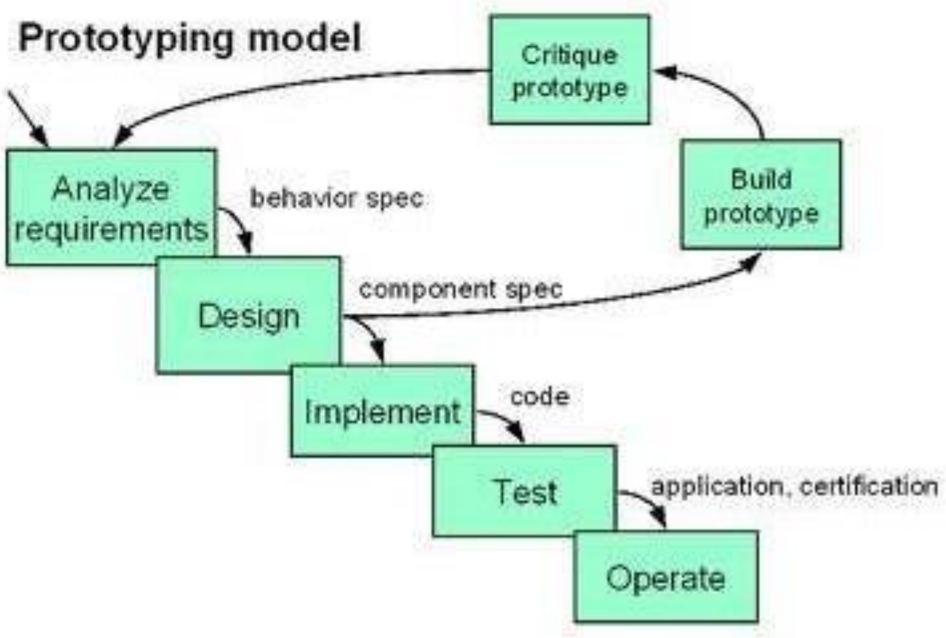

Figure 5: Software Prototyping Model [CITATION Leh15 \l 3081]

Looking at the advantages of this SDM like finding the missing functionality in the system before completion and also gets the better understanding the system before the final product. So, considering these advantages this developer had considered this methodology, but doing further research about the system, the developer had come to the conclusion that this SDM cannot be used as it was not suitable for this project. As there is a time constraint while doing this project and it requires more time and effort to develop prototype only and the project may run out of time developing prototype only. Another problem was considered was this methodology, it will increase the complexity as the system may go beyond the original plan. Finally, this SDM is not chosen.

3.1.2 Big Bang Model

The Big Bang model is another SDM in which does not follow any particular process. In this SDM planning is also not given any priority, whereas the majority of time is spent on development phase and software is developed.

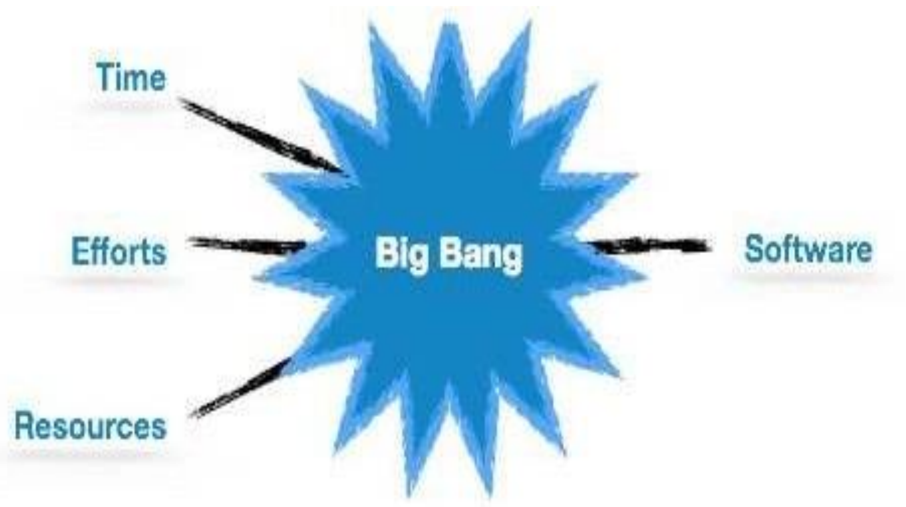

Figure 6: Big Bang Model [CITATION Tut15 \l 3081]

This methodology was considered by the developer as there are advantages like it is simple to carry out, where managing is easy and lots of flexibility is given to developer as they should not follow any specific order. Along with these this SDM also requires very little resources comparison to other SDM. Although it has some advantages, but developer finds it is not suitable for this project as it has disadvantages like there is not any planning while doing this and lots of chances of failure of the project. The developer has also done research on the project and is well known about the project so there is a good plan to make the project. So, reviewing the advantages and disadvantages of the Big Bang Model it is not chosen.

3.1.3 Incremental Model

Incremental Model is developed with the improvement of the waterfall model. In the incremental model, the project is broken down into the different standalone component of software development lifecycle. Those cycles are also broken to smaller increment which is easy to manage. [CITATION IST14 \l 3081]

Incremental Model was considered as project requirement was clear and is also easy to break software and there was not any overlap between the different increment. As this SDM is very flexible as a developer can plan to make multiple plans at once and also like in changing requirement and scope can be done during development also. Not only this error can be also easily found in this SDM which help to minimize the risk. The developer has also personally felt that breaking this project into different increment will make easy to complete this project. So, this model is chosen.

3.2 Selected Methodology and Phases

Finally, after considering different methodologies Incremental Model is chosen for this project. While completing the project with the incremental model the project is divided into different increment like system 1 will be work related to light and motion and system 2 will be related to controlling water. According to Incremental Model system 1 will be completed and then system 2 will be started. The phases of Incremental Model are briefly described below along with work that will be conducted in this project.

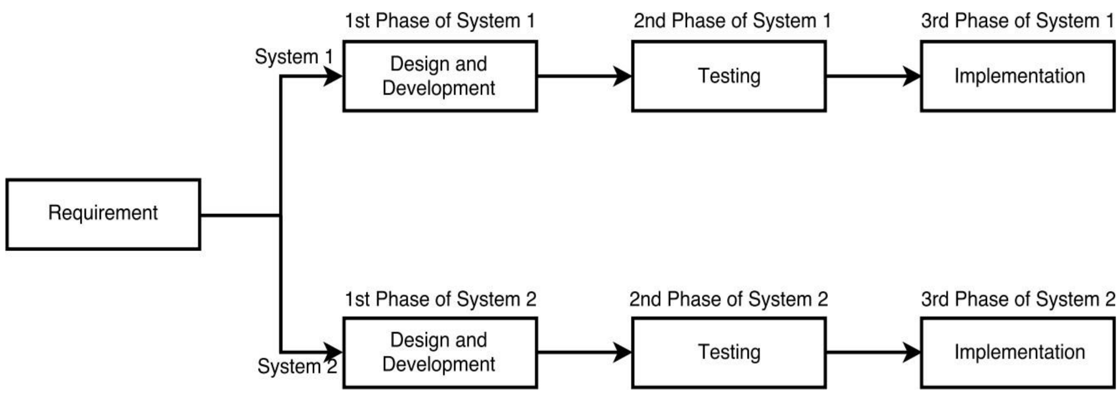

Figure 7: Incremental Model

a. Requirement phase

Requirement phase is the first phase of the incremental model where requirements are identified by the system. In this phase developer will do research about the IoT and Home Automation of System and their usability along with the things that can be completed in this project will be analyzed. In this phase, the developer will find out the complete functional requirement and analysis of all systems that will be included in this project.

b. Design and Development phase

In this phase of the incremental model, the system and design of the system is prepared by analyzing from the requirements. The developer will develop the lighting system in phase 1 of system 1 and then the developer will develop a water sensor system in this phase of different increment. The developer will develop the new functionality of the system in each of the new phases as per the incremental model.

c. Testing phase

After design and development is completed in the incremental model, then there is testing phase. In this phase test all the existing functions as well as non-functional testing is done by the developer. The developer will test the behaviour of the system and its response as per the requirement. Testing is carried out differently as per the number of increment done.

d. Implementation phase:

After completion of testing, then the system is implemented. In this phase developer will finalize every part of the system if there is any error or anything need to be added and will make ready to use. Users can also use the system after the implementation phase. After the implementation of the system 1 and then will start to make system 2 as per the plan for now.

3.3 Engagement

In this project, requirement phase is completed and system 1 is under being developed. In the development phase, the project is in the phase of system 1 design and development. As requirement phase is completed so all the functional and non-functional are completely identified. The project is going per plan for now and if it goes as per the plan, then design and development will be completed within a few weeks and testing phase of the system will be started.

4. Progress

4.1 Work Completed

Tasks	**Finished Date**	**Development**	**Reason**
Selecting the topic	31/10/2021	Completed	
Research about topic	05/11/2021	Completed	
Topic Selected	10/11/2021	Completed	
Feasibility Study	13/11/2021	Completed	
Analysis of Study	14/11/2021	Completed	
Research on topic	19/10/2021	Completed	
Looking similar project	27/10/2021	Completed	
Looking Methodologies	28/10/2021	Completed	
Preparing Questionnaire	01/11/2021	Completed	
Survey	04/11/2021	Completed	
Analysis of Survey	04/11/2021	Complete	
Report Preparation	09/11/2021	Complete	
Report Finalisation	14/11/2021	Complete	

Table 2: Current Progress

As project is almost mid of its completion phase. At this mid phase almost half of the work is completed. The description of current completed work of this project is defined. The developer has completed all the work almost in the time. There has not been problem with completing work on time.

4.2 Future Work

Tasks	Anticipated Finished Date
Design and Development of system 1	10/1/2022
Design and Development of system 2	03/08/2022
Merging the system	20/03/2022
Survey about system	23/03/2022
Survey about Project	14/04/2022
Deploying the system	26/04/2022
Preparing the documentation	26/04/2022
Finalising the documentation	28/04/2022

Table 3: Further Work

Likewise till now if project goes as per plan then the project is expected to be completed on 15th February, 2022. There is not any problem faced till now in this project.

4.3 Progress Review

Tasks	**Finished Date**	**Development**	**Reason**
Selecting the topic	31/10/2021	Completed	
Research about topic	05/11/2021	Completed	
Topic Selected	10/11/2021	Completed	
Feasibility Study	13/11/2021	Completed	
Analysis of Study	14/11/2021	Completed	
Research on topic	19/10/2021	Completed	
Looking similar project	27/10/2021	Completed	
Looking Methodologies	28/10/2021	Completed	
Preparing Questionnaire	01/11/2021	Completed	
Survey	04/11/2021	Completed	
Analysis of Survey	04/11/2021	Completed	
Report Preparation	09/11/2021	Completed	
Report Finalisation	14/11/2021	Completed	
Design and Development of system 1	10/01/2022	Progress	Developer is still on this phase and process of design and development is going on.
Design and Development of system 2	11/01/2022	Incomplete	
Merging the system	20/03/2022	Incomplete	
Survey about system	23/03/2022	Incomplete	
Survey about Project	14/04/2022	Incomplete	
Deploying the system	26/04/2022	Incomplete	
Preparing the documentation	26/04/2022	Incomplete	
Finalising the documentation	28/04/2022	Incomplete	

Table 4: Progress Review

Progress is still going on of the project without delay.

4.4 Work Breakdown Structure

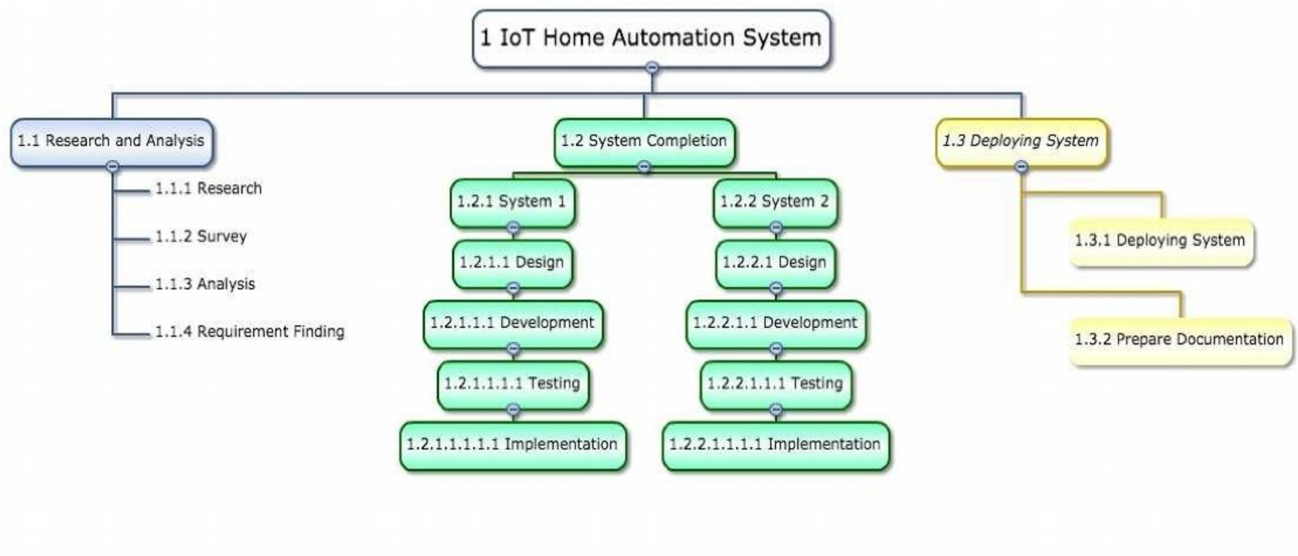

Figure 8: Work Breakdown Structure

Work is divided into different component to make work easier. This will help to track the work and will also help to make plan in future.

4.5 Gantt Chart

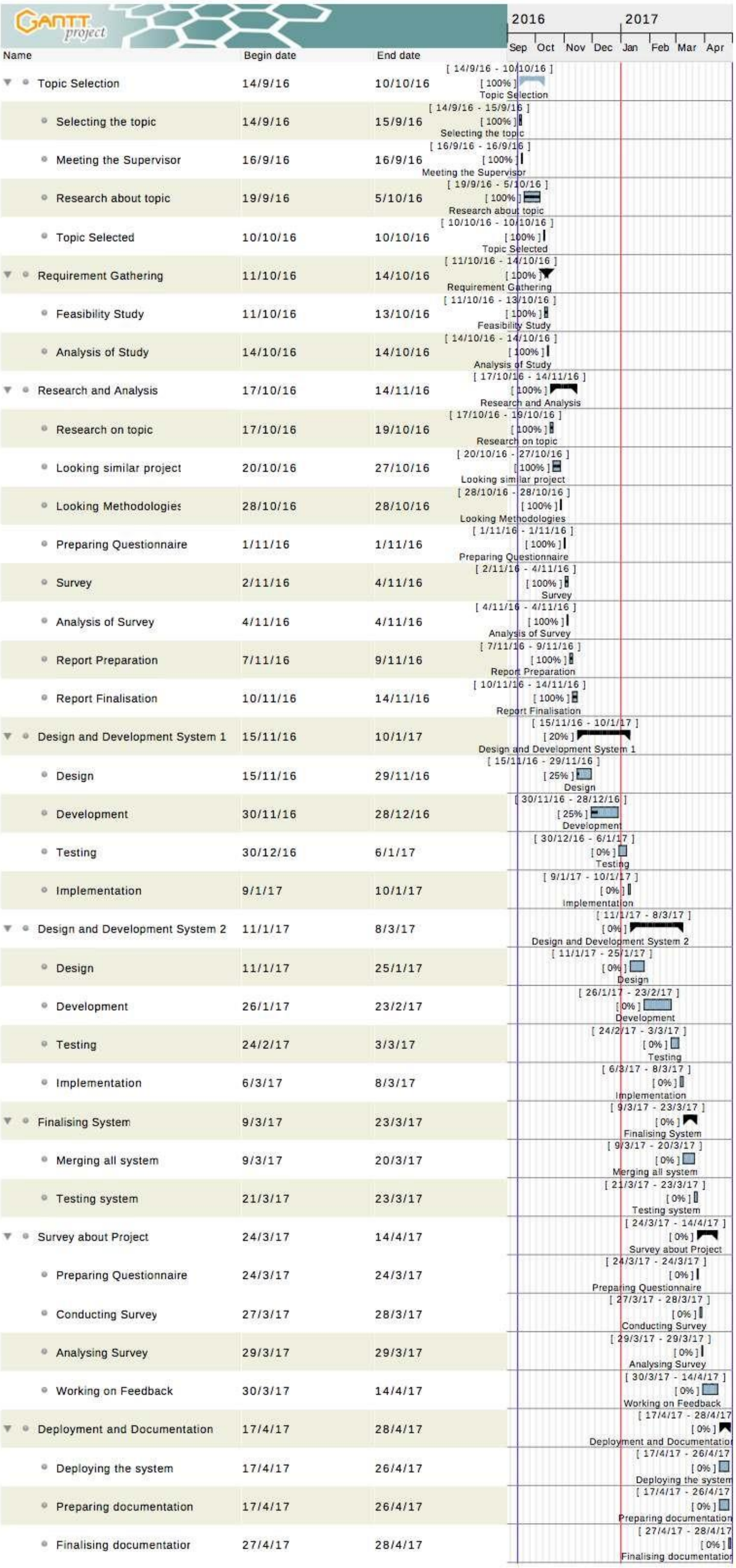

Figure 9: Gantt Chart (32% of work is completed)

5. Conclusion

Home Automation is a big project and can be extended to any level in the present scenario. All the manual work carried out in our home currently can be atomized to some level. So, due to time constraint with limited resources developer has narrowed this project with two major features for Home Automation. A further enhancement that will be carried out in this project is described as Automatic Lightning System and Automatic Water Control.

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8. Appendix

8.1 Annex A: Survey

Questions:

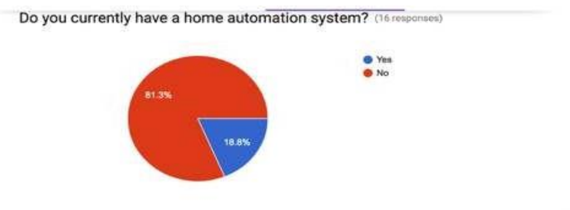
1. Do you currently have a home automation system?
2. Are you interested in home automation system?
3. If yes, what automation system you would consider at first to use?
4. If no, for any specific reason?
5. For no, would you use home automation if your problem is solved?

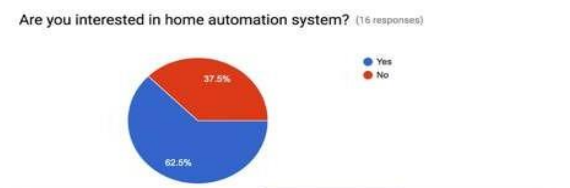
Objective of Survey: Find out the user of home automation and analyse will they use home automation in future.

Target Group: Early adaptor of Technology

Timeframe: 2 days

Result of Survey:



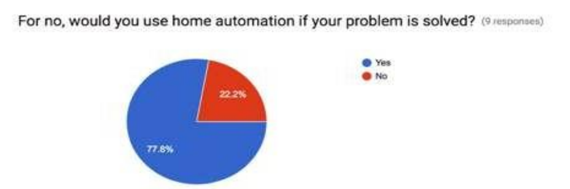


If yes, what automation system you would consider at first to use? (9 responses)

security system in door
Know about my house water system
all thing in my house should be automated
automated light at first
seems useless
automated light and security
light and gas
automated light and water
automatic light at night and water motor

If no, for any specific reason? (3 responses)

expensive
it should be reliable
cost is too high



8.2 Annex 2: Functional and Non-functional of the project

a. Functional Requirement:

- Automation Lighting System
- Automation Water Sensor System
- Web Application

b. Non-Functional Requirement:

- Usability
- Durability
- Reliability
- Maintainable
- Integrity
- Simple
- Efficient